MIL-E-1/812C
22 May 1975
SUPERSEDING
MIL-E-1/812B
20 October 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 5D22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode

See figure 1

Mounting position: Vertical, base up or down

Weight: 8-ounces (226.8 grams) nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec1	Ec2	Ib	Pp	Pg1	Pg2	Cooling
Unit:	MHz	Vac	Vdc	Vdc	Vdc	mAdc	W	W	W	Note 1
Maximum:	75	5.25	4,000	-500	600	350	250	10	35	---
Minimum:	---	4.75	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	5.0	2,500	Adjust	500	100	---	---	---	Note 2

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	13.5	14.7	Aac
1266	Total grid current	3		0.65	II	Ic1	---	-10	µAdc
1266	Primary grid emission (control)	-	Ef = 5.5 Vac; Ic1 = 165 mAdc; t = 15 sec; anode and screen grid floating	0.65	II	Isg1	---	-100	µAdc
1266	Primary grid emission (screen)	-	Ef = 5.5 Vac; Ec1 = 0; Ic2 = 135 mAdc; t = 15 sec; anode floating	0.65	II	Isg2	---	-200	µAdc
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-65	-95	Vdc
1231	Peak emission	-	eb = ec1 = ec2 = 2,500 v	0.65	II	is	5.0	---	a
	Quality conformance inspection, part 2								
1316	Amplification factor	-	g1 to g2; Ec2 = 500 Vdc; Ic2 = 70 mAdc; Eb = 0 (anode grounded)	---	---	Mu	4.5	6.0	---
1236	Power oscillation (1)	-	Ebb = 3,000 Vdc; Ib = 200 mAdc; Ec2 = 350 Vdc; Rg1 = 12,000 ohms; F = 15 MHz	---	---	Po	350	---	W (useful)
				---	---	Ic2	50	100	mAdc
1331	Direct-interelectrode capacitance			---	---	Cgp	---	0.14	pF
						Cin	10.7	14.5	pF
						Cout	3.7	5.1	pF
	Quality conformance inspection, part 3								
Ⓒ 1031	Low-frequency vibration	4	No voltages applied	---	---	---	---	---	---
1036	Bump	4	Angle = 20°	---	---	---	---	---	---
Ⓒ ---	Low-frequency vibration and shock end points:								
1261	Electrode voltage (grid)	-		---	---	Ec1	-65	-95	Vdc
1266	Total grid current	-		---	---	Ic1	---	-10	µAdc
---	Life test	-	Group C; power oscillation (1); t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission	-		---	---	is	4.0	---	a
1266	Primary grid emission (control)	-		---	---	Isg1	---	-100	µAdc
1266	Primary grid emission (screen)	-		---	---	Isg2	---	-200	µAdc
1266	Total grid current	-		---	---	Ic1	---	-10	µAdc
1236	Power oscillation (2)	4	Ebb = 3,000 Vdc; Ib = 200 mAdc; Ec2 = 350 Vdc; Rg1 = 12,000 ohms; F = 75 MHz	---	---	Po	350	---	W (useful)

NOTES:

1. Adequate cooling shall be provided for the seals and envelope of the tube. An air-system socket and air director (chimney) is desirable, such as Eimac SK-400/SK-406 or SK-410/SK-406, or equal. At frequencies below 30 MHz, cooling air in the amount of 5 cfm around the base of the tube past the envelope is required. At frequencies above 30 MHz, losses in the leads and envelope contribute to heating, and additional cooling may be required. Cooling should be applied simultaneously with filament power, and in all cases there should be some movement of air past the base and envelope.

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NOTES: -Continued

2. Cooling of the tube is permitted to an extent not to exceed that specified in note 1.

3. This test shall be the first test performed at the conclusion of the holding period.

© 4. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force -17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

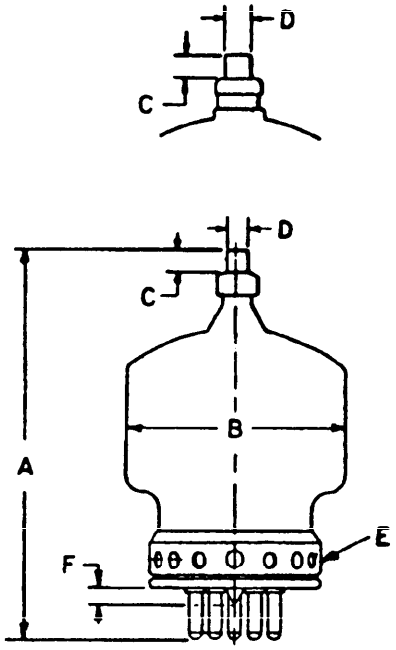
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2937)

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PIN CONNECTIONS	
Pin No.	Element
1	f
2	g2
3	g1
4	g2
5	f
cap	a



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.875 (149.23)	6.375 (161.93)
B	---	3.563 (90.50)
F	---	.250 (6.35)
Ⓒ Quality conformance inspection, part 3 (see note)		
C	.328 (8.33)	---
D	.350 (8.89)	.365 (9.27)
E	A5-97 (EIA)	

NOTE:
These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total allowable failures
n2 = 4 c2 = 1; for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3, basis.

FIGURE 1. Outline drawing of electron tube type 5D22.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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